

MPG No. 13.3.46  $4'm'm$  (4'm'm setting) [ Type III, tetragonal ] [M tensor]

\* Rank 0 tensor. \* Rank 1 tensor. \* Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_v^{(1)}$$

\* Rank 2 tensor (a). \* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ 0 & -M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{\beta(1)}$$

\* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ 0 & \frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{T}_{xy}^{(1)}$$

$$\frac{M_{xxz}}{2} = \mathbb{T}_{xy}^{(1)}$$

\* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ M_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzy} = \mathbb{T}_{xy}^{(2)}$$

\* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & 0 \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxzz} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)}$$

$$M_{yzyz} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)}$$

\* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & 0 \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = -\mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

\* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2\mathbb{M}_v^{(4)}$$

\* Rank 4 tensor (aaa). \* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

\* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

\* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzzz} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{zzxx} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{yzyz} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzzz} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

\* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzz} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzzz} &= -2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{xyxy} &= 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}
\end{aligned}$$

\* Rank 4 tensor (t).

$$\begin{bmatrix}
M_{xxxx} & 0 & 0 \\
0 & -M_{xxxx} & 0 \\
0 & 0 & 0 \\
0 & 0 & M_{yyzz} \\
M_{zzxx} & 0 & 0 \\
0 & -M_{yyzz} - M_{zzxx} & 0 \\
0 & M_{yzzz} & 0 \\
0 & 0 & M_{zzxz} \\
-M_{yzzz} - M_{zzxz} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)} \\
M_{yyzz} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{zzxx} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{yzzz} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\
M_{zzxz} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)}
\end{aligned}$$